

Objective: Analyzing network traffic to detect security threats (no anomalies found)

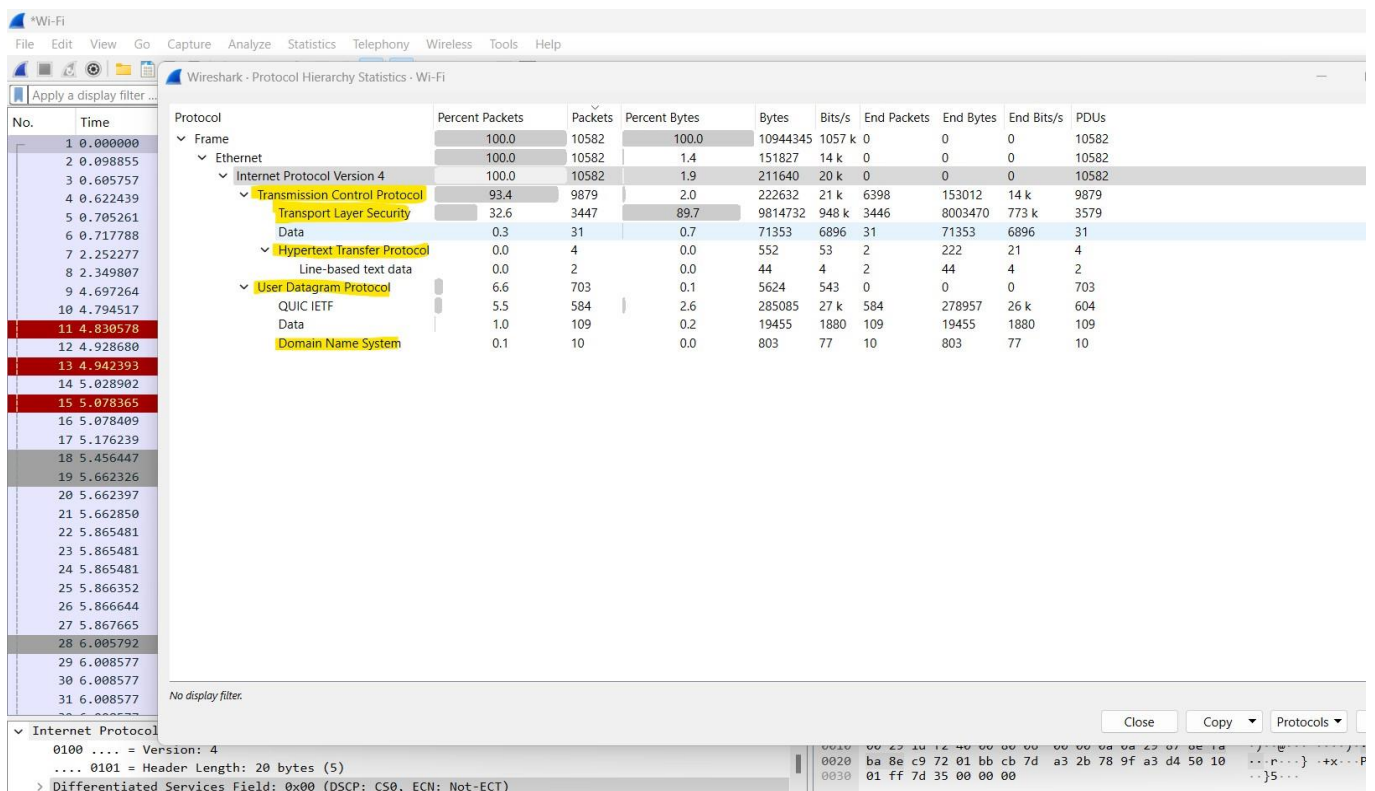
Date of Analysis: 21.01.2025, 13:22

Tools Used:

- Wireshark
- VirusTotal (for threat intelligence)

Network Traffic Overview

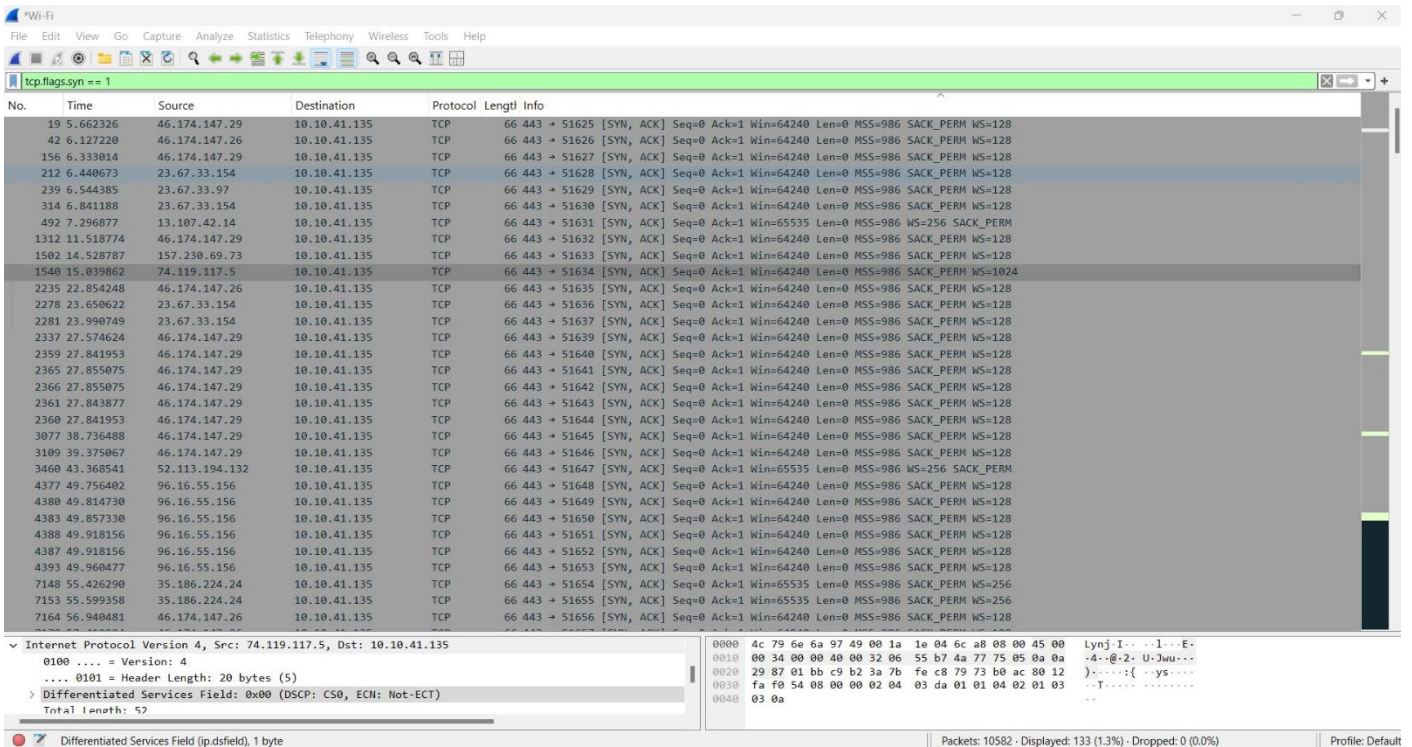
- Total Packets Captured: 10582
- Common Protocols Detected: HTTP, DNS, TLS, UDP & TCP



- Capture duration: 5 minutes
- Test website: www.emag.ro

Filters Used

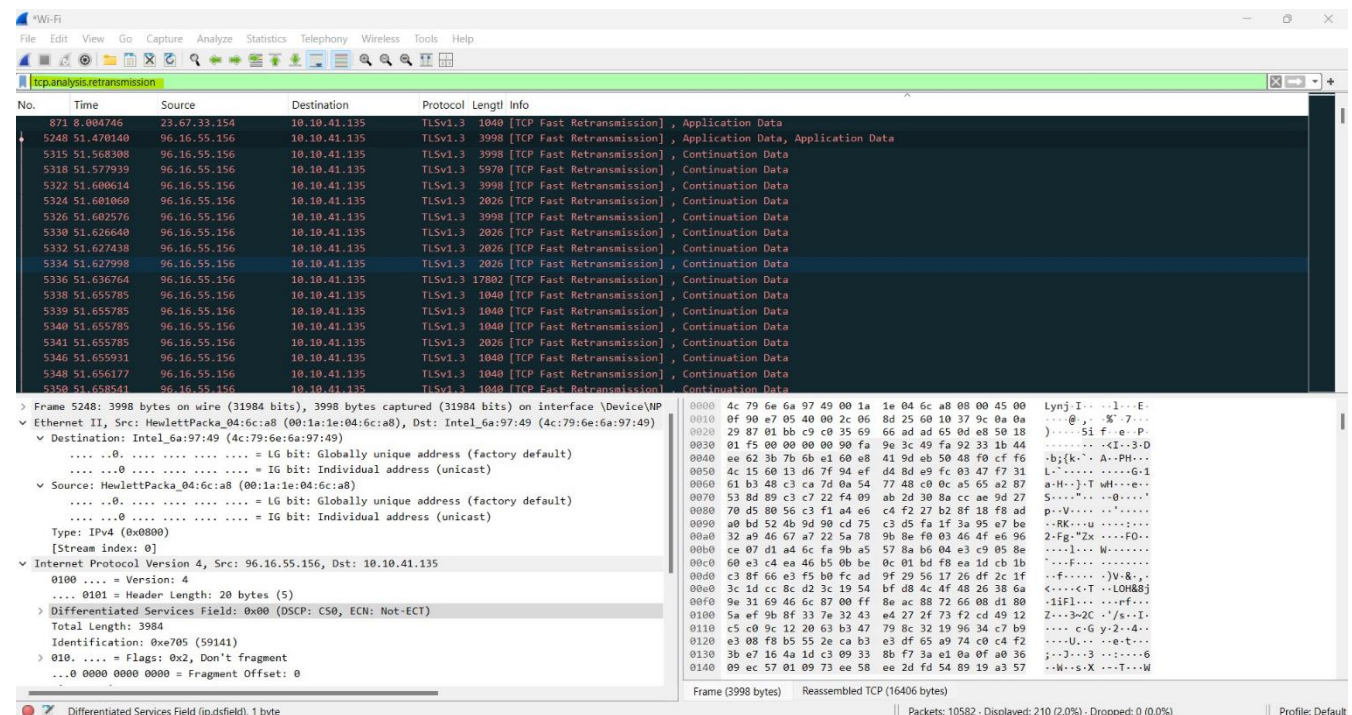
- SYN Packets: the reason was to monitor TCP handshakes and to find potential issues in the connection initiation phase.



The image shows a Wireshark capture of network traffic. The top pane displays a list of packets, with a filter applied: `tcp.flags.syn == 1`. The middle pane shows the details of the selected packet (No. 19), including the Ethernet II header, Internet Protocol Version 4 header, and the Hypertext Transfer Protocol header. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
19	5.662326	46.174.147.29	10.10.41.135	TCP	66	443 → 51625 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
42	6.127220	46.174.147.29	10.10.41.135	TCP	66	443 → 51626 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
156	6.333814	46.174.147.29	10.10.41.135	TCP	66	443 → 51627 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
212	6.440673	23.67.33.154	10.10.41.135	TCP	66	443 → 51628 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
239	6.544385	23.67.33.97	10.10.41.135	TCP	66	443 → 51629 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
314	6.841188	23.67.33.154	10.10.41.135	TCP	66	443 → 51630 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
492	7.296877	13.187.42.14	10.10.41.135	TCP	66	443 → 51631 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=986 WS=256 SACK_PERM
1312	11.518774	46.174.147.29	10.10.41.135	TCP	66	443 → 51632 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
1502	14.528787	157.230.69.73	10.10.41.135	TCP	66	443 → 51633 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
1540	15.039862	74.119.117.5	10.10.41.135	TCP	66	443 → 51634 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=1024
2235	22.854248	46.174.147.26	10.10.41.135	TCP	66	443 → 51635 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
2278	23.650622	23.67.33.154	10.10.41.135	TCP	66	443 → 51636 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
2281	23.980749	23.67.33.154	10.10.41.135	TCP	66	443 → 51637 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
2337	27.574624	46.174.147.29	10.10.41.135	TCP	66	443 → 51639 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
2359	27.841953	46.174.147.29	10.10.41.135	TCP	66	443 → 51640 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
2365	27.855075	46.174.147.29	10.10.41.135	TCP	66	443 → 51641 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
2366	27.855075	46.174.147.29	10.10.41.135	TCP	66	443 → 51642 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
2361	27.843877	46.174.147.29	10.10.41.135	TCP	66	443 → 51643 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
2360	27.841953	46.174.147.29	10.10.41.135	TCP	66	443 → 51644 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
3077	38.736488	46.174.147.29	10.10.41.135	TCP	66	443 → 51645 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
3109	39.375067	46.174.147.29	10.10.41.135	TCP	66	443 → 51646 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
3460	43.368541	52.113.194.132	10.10.41.135	TCP	66	443 → 51647 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=986 WS=256 SACK_PERM
4377	49.756402	96.16.55.156	10.10.41.135	TCP	66	443 → 51648 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
4380	49.814730	96.16.55.156	10.10.41.135	TCP	66	443 → 51649 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
4383	49.857330	96.16.55.156	10.10.41.135	TCP	66	443 → 51650 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
4388	49.918156	96.16.55.156	10.10.41.135	TCP	66	443 → 51651 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
4387	49.918156	96.16.55.156	10.10.41.135	TCP	66	443 → 51652 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
4393	49.960477	96.16.55.156	10.10.41.135	TCP	66	443 → 51653 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128
7148	55.426290	35.186.224.24	10.10.41.135	TCP	66	443 → 51654 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=986 SACK_PERM WS=256
7153	55.599358	35.186.224.24	10.10.41.135	TCP	66	443 → 51655 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=986 SACK_PERM WS=256
7164	56.940481	46.174.147.26	10.10.41.135	TCP	66	443 → 51656 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=986 SACK_PERM WS=128

- Dropped or Out-of-Order Packets: Checked for packet drops or reordering that could indicate problems responsible for a slower or a less reliable network.



The image shows a Wireshark capture of network traffic. The top pane displays a list of packets, with a filter applied: `tcp.analysis.retransmission`. The middle pane shows the details of the selected packet (No. 871), including the Ethernet II header, Internet Protocol Version 4 header, and the Hypertext Transfer Protocol header. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
871	8.004746	23.67.33.154	10.10.41.135	TLSv1.3	1040	[TCP Fast Retransmission], Application Data
5248	51.470140	96.16.55.156	10.10.41.135	TLSv1.3	3998	[TCP Fast Retransmission], Application Data
5315	51.568308	96.16.55.156	10.10.41.135	TLSv1.3	3998	[TCP Fast Retransmission], Continuation Data
5318	51.577939	96.16.55.156	10.10.41.135	TLSv1.3	5970	[TCP Fast Retransmission], Continuation Data
5322	51.600614	96.16.55.156	10.10.41.135	TLSv1.3	3998	[TCP Fast Retransmission], Continuation Data
5324	51.601060	96.16.55.156	10.10.41.135	TLSv1.3	2826	[TCP Fast Retransmission], Continuation Data
5326	51.602576	96.16.55.156	10.10.41.135	TLSv1.3	3998	[TCP Fast Retransmission], Continuation Data
5330	51.626640	96.16.55.156	10.10.41.135	TLSv1.3	2826	[TCP Fast Retransmission], Continuation Data
5332	51.627438	96.16.55.156	10.10.41.135	TLSv1.3	2826	[TCP Fast Retransmission], Continuation Data
5334	51.627998	96.16.55.156	10.10.41.135	TLSv1.3	2826	[TCP Fast Retransmission], Continuation Data
5336	51.627664	96.16.55.156	10.10.41.135	TLSv1.3	17802	[TCP Fast Retransmission], Continuation Data
5338	51.655785	96.16.55.156	10.10.41.135	TLSv1.3	1040	[TCP Fast Retransmission], Continuation Data
5339	51.655785	96.16.55.156	10.10.41.135	TLSv1.3	1040	[TCP Fast Retransmission], Continuation Data
5340	51.655785	96.16.55.156	10.10.41.135	TLSv1.3	1040	[TCP Fast Retransmission], Continuation Data
5341	51.655785	96.16.55.156	10.10.41.135	TLSv1.3	2926	[TCP Fast Retransmission], Continuation Data
5346	51.655931	96.16.55.156	10.10.41.135	TLSv1.3	1040	[TCP Fast Retransmission], Continuation Data
5348	51.656177	96.16.55.156	10.10.41.135	TLSv1.3	1040	[TCP Fast Retransmission], Continuation Data
5350	51.658541	96.16.55.156	10.10.41.135	TLSv1.3	1040	[TCP Fast Retransmission], Continuation Data

- TCP Handshakes and Connections: In order to detect abnormal connection attempts or interruptions.

Wireshark packet capture showing a list of TCP SYN-ACK packets. The packet list pane shows a series of packets from 19 to 7164, all with the same source and destination IP addresses (10.10.41.135). The packet details pane shows the structure of a TCP packet, including the header and options. The packet bytes pane shows the raw data of the packet.

- Presence of larger packets – filter used to expose the presence of large data transfers or potential issues. Even though I could find many large packets (e.g: 3998 bytes), the source IP is familiar and follows the usual network behavior, which leads to the conclusion that this packet is simply normal traffic.

Wireshark packet capture showing a large packet (3998 bytes) from source IP 10.10.41.135 to destination IP 10.10.41.135. The packet details pane shows the structure of the packet, including the header and options. The packet bytes pane shows the raw data of the packet.

Wireshark · Packet 353 · Wi-Fi

> Frame 353: 3998 bytes on wire (31984 bits), 3998 bytes captured (31984 bits) on interface \Device\NPF_{9389C5CF-7D41-431F-B951-A3913350D8A4}, id 0

> Ethernet II, Src: HewlettPacka_04:6c:a8 (00:1a:1e:04:6c:a8), Dst: Intel_6a:97:49 (4c:79:6e:6a:97:49)

✓ Internet Protocol Version 4, Src: 23.67.33.154, Dst: 10.10.41.135

0100 = Version: 4
 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total length: 3984
 Identification: 0xe5d8 (58840)

> 010. = Flags: 0x2, Don't fragment
 ...0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 43
 Protocol: TCP (6)
 Header Checksum: 0xee21 [validation disabled]
 (Header checksum not shown, because it was not verified)

```

0000  4c 79 6e 6a 97 49 00 1a 1e 04 6c a8 08 00 45 00  Lynj.I...l...E-
0010  0f 90 e5 d8 40 00 2b 06 ee 21 17 43 21 9a 0a 0a  ...@+...!C!...
0020  29 87 01 bb c9 ac 27 11 3d 2d 8a d0 86 11 50 18  )....' =....P.
0030  01 f5 00 00 00 00 1b 77 88 44 b3 b8 b9 d1 84 f2  ....w-D.....
0040  74 58 95 a2 cc 1d 68 56 70 87 6c 34 3a 86 4d 8e  tX...hV p.14..M.
0050  8e 8f 05 43 0b 4a 1c 93 27 9c 7b e0 2d 5e d6 b3  ...C.J... '{-^...
0060  55 3c c4 bc 8a 04 f4 0f 8a e6 00 b8 3a 1f ff 17  U<.....
0070  c6 c4 72 83 9b 3d a6 0f 9e ab 96 82 83 67 4e 7a  ..r...gNz
0080  32 5d 12 0a d2 59 ad 2d 8d ff 63 d6 d9 1e a6 5c  2]...Y...c....\
0090  e1 cf ed 65 7f 53 1a 19 aa fa 83 32 67 96 8e 6d  ...e.S...2g.m
00a0  57 02 61 aa 07 ee 67 b1 92 eb 25 21 83 07 b7 b6  W.a...g...%!...
00b0  2f 98 c5 18 9a aa d6 e0 f2 82 69 a6 83 6b 89 d0  /. ....i..k...
00c0  d3 35 e2 be 3c 05 bf 63 42 44 67 23 d0 46 67 59  .5...c BDg#.FgY
00d0  d7 d4 3e fa a9 1a 33 f9 dd 64 f7 b1 59 94 d6 d6  ..>...3..d..Y...
00e0  77 e1 30 ab 3b 2a 23 5a 2c ba d2 12 69 fc 7b b3  w.0;.*#Z ,...i.{.
00f0  e7 99 49 dd 5f 2d 62 e8 67 4e 48 7f 37 23 26 b4  ..I.-b. gNH-7#&.
0100  83 6c 54 0b 50 1f 88 51 7a 03 6f b0 65 a8 ec 2b  .lT.P..Q z.o.e.+
0110  52 ab d6 3f 24 22 b8 73 69 a4 9d 1e 50 0c 66 84  R..?$.s i...P.f.
  
```

No.: 353 · Time: 6.958215 · Source: 23.67.33.154 · Destination: 10.10.41.135 · Protocol: TCP · Length: 3998 · Info: 443 → 51628 [PSH, ACK] Seq=10442 Ack=3025 Win=64128 Len=3944 (TCP PDU reassembled in 358)

☒ Show packet bytes Layout: Vertical (Stacked)

- **SSH Brute-Force Detection** - Applied this filter to detect any unusual SSH connection attempts indicating brute-force attacks.

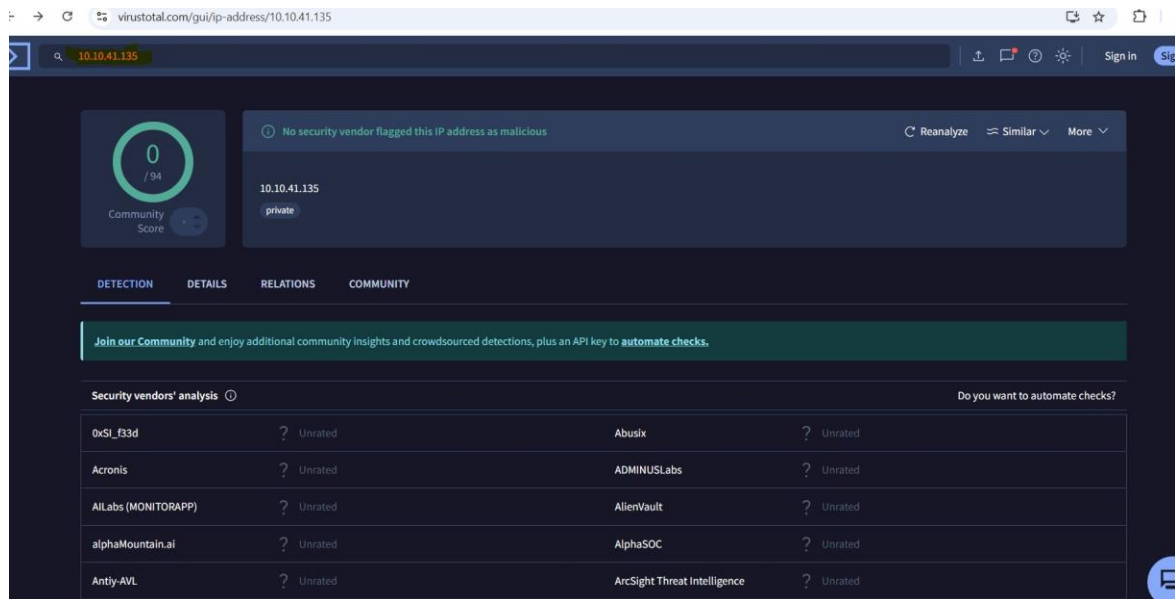
Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 22 and tcp.flags.syn == 1

No.	Time	Source	Destination	Protocol	Length	Info

- Virus Check for IP: Checked for signs of malware using Virustotal.com of the destination IP address 10.10.41.135.



Results:

No Anomalies Found: After applying the above mentioned filters, I could observe no suspicious behavior or anomalies.

The attached screenshots can clearly state that no issues were identified during the network analysis.

Further recommendations:

- maintaining the current configuration of network devices, DNS servers, and security tools.
- regular monitoring of network traffic to establish baseline metrics; this way if the traffic volume or patterns deviate in the future, it will be easier to detect anomalies.
- proactive monitoring with a focus on DNS.